Adopted Levels:tentative

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh	NDS 156, 70 (2019)	31-Jan-2019

Q(β^-)=1970 *CA*; S(n)=7280 *CA*; S(p)=2960 *CA*; Q(α)=10970 *CA* 1997Mo25
S(2n)=13200, S(2p)=4930 (1997Mo25,theory).

2017Ka66: tentative assignment to ^{294}Lv in ^{248}Cm(^{48}Ca,2n),E=261.6 MeV (E^*=41.3 MeV reaction using RIKEN Linear Accelerator (RILAC) and gas-filled recoil ion separator (GARIS). One EVR-α-α-SF correlated event (chain #4) was interpreted in three ways, one being ^{294}Lv → ^{290}Fl → ^{286}Cn decay chain, while the other two involved (^{48}Ca,3n) channel producing ^{293}Lv → ^{289}Fl → ^{285}Cn, and ^{293}Lv → ^{289}Fl → ^{285}Cn → ^{281}Ds chains.

History of Chain #4 (2017Ka66):
Chain #4 (first interpretation): Energy of EVR=19.0 MeV.
$E_{\alpha 1}$ missing α, Δt_1=?, assigned to ^{293}Lv.
$E_{\alpha 2}$=9.72 MeV *3*,Δt_2=0.66 s-τ(^{293}Lv), assigned to ^{289}Fl.
$E_{\alpha 3}$ missing α, Δt_3 ?, assigned to ^{285}Cn.
E_{SF}=221 MeV, Δt_4=12.19−τ(^{285}Cn s, assigned to ^{281}Ds: SF decay.
Chain #4 (second interpretation): Energy of EVR=19.0 MeV.
$E_{\alpha 1}$ missing α, Δt_1=?, assigned to ^{293}Lv.
$E_{\alpha 2}$=9.72 MeV *3*,Δt_2=0.66 s-τ(^{293}Lv), assigned to ^{289}Fl.
E_{SF}=221 MeV, Δt_3=12.19 s, assigned to ^{285}Cn: SF decay.
Chain #4 (third interpretation): Energy of EVR=19.0 MeV.
$E_{\alpha 1}$ not observed, Δt_1=?, assigned to ^{294}Lv.
$E_{\alpha 2}$=9.72 MeV *3*,Δt_2=0.66 s-τ(^{294}Lv), assigned to ^{290}Fl.
E_{SF}=221 MeV, Δt_3=12.19 s, assigned to ^{286}Cn: SF decay.

^{294}Lv Levels

E(level)	J^π	Comments
0?	0^+	α decay mode is most likely, as theoretical $T_{1/2}(\alpha)$=18 ms and $T_{1/2}(\beta)$>100 s (1997Mo25).